

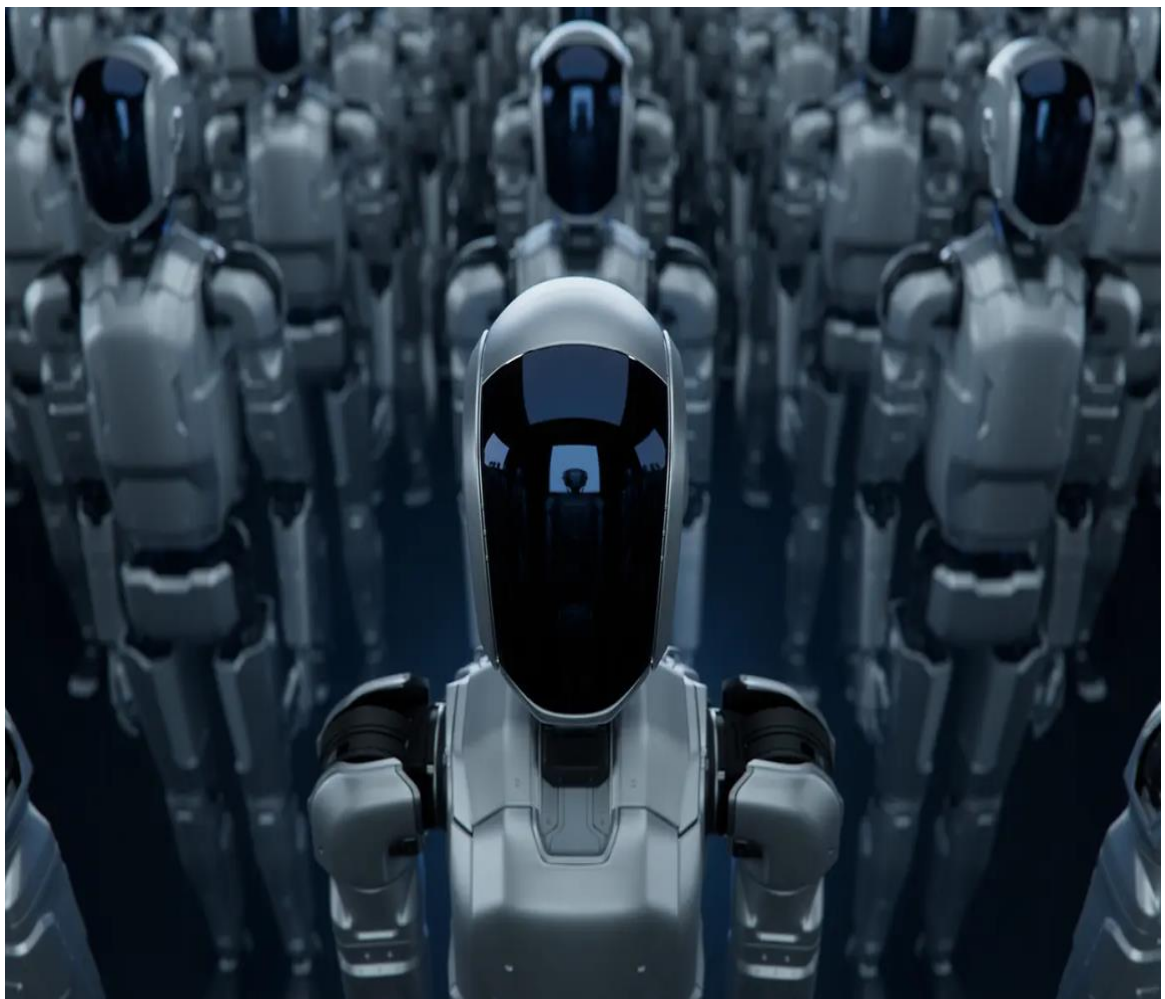
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

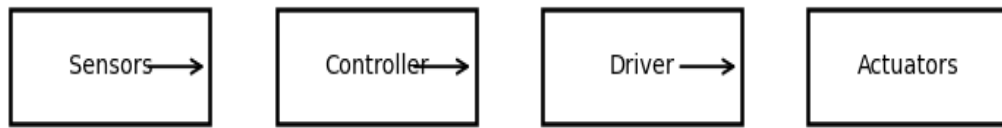
TOPIC: ROBOTICS

Prepared By: Shinde Kalpana Tanaji

ROBOTICS

- ROBOTICS

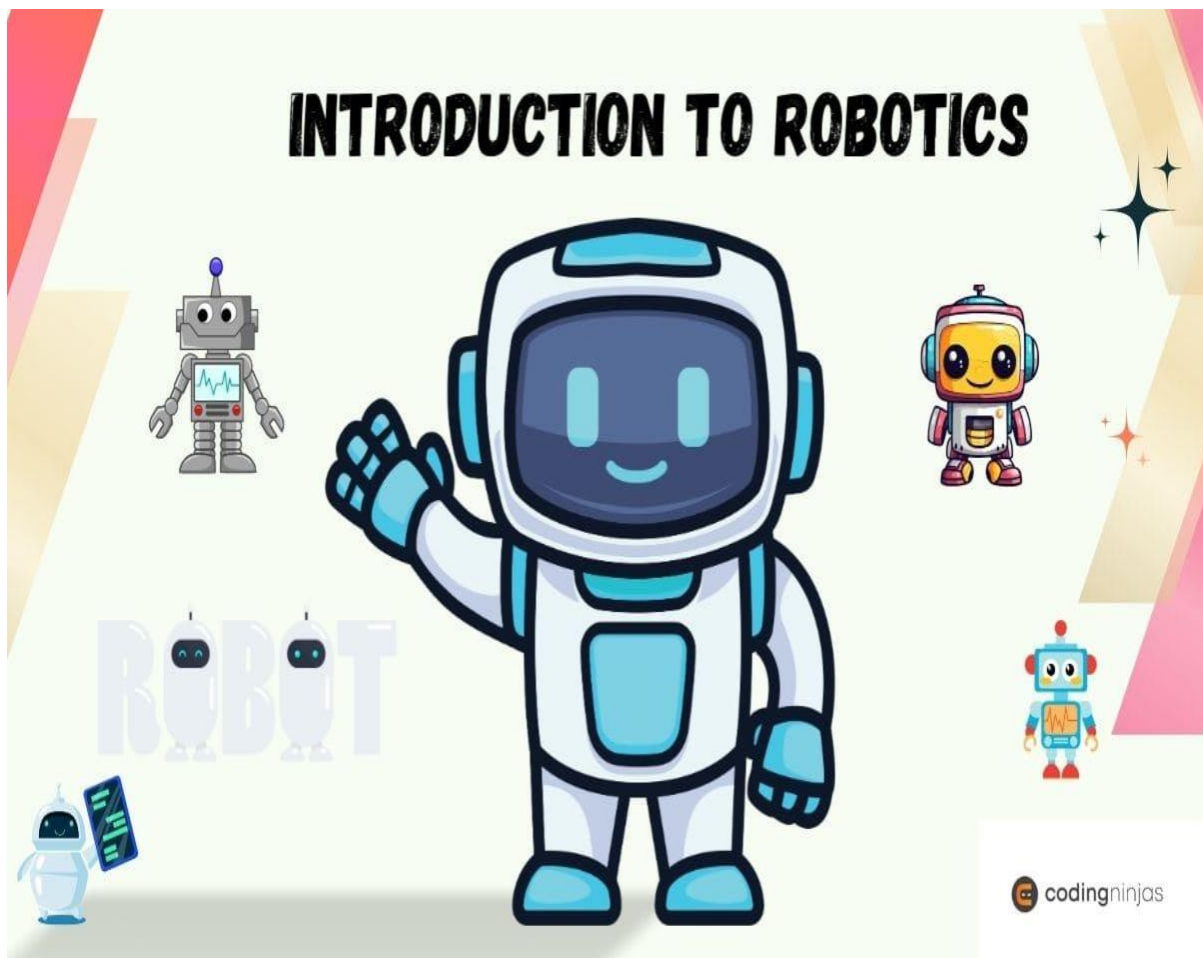




ROBOTICS

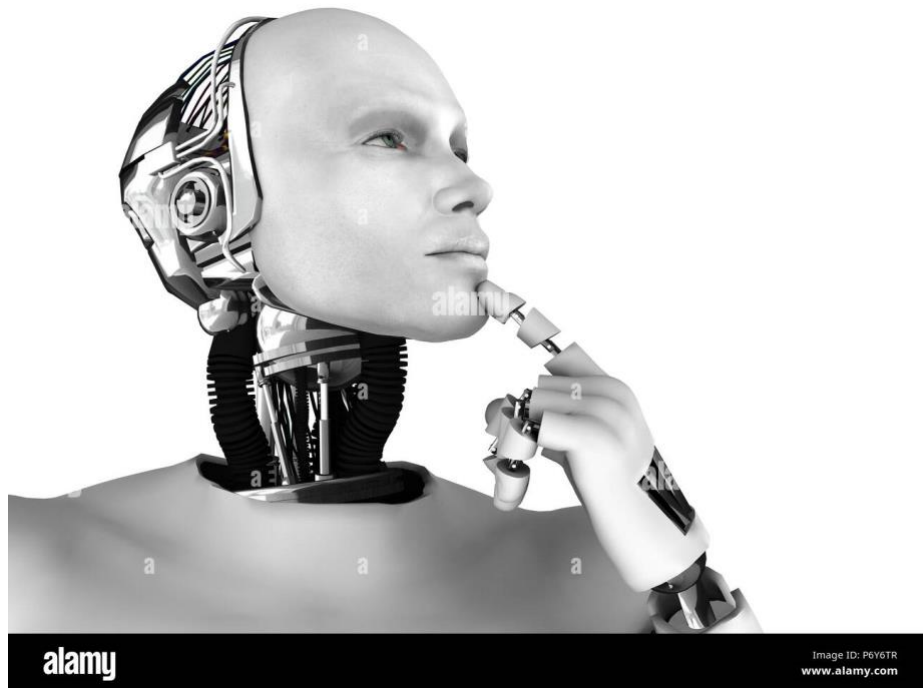
➤ A Beginner-Friendly Introduction to Robots and Robotic Systems.

- **Introduction to Robotics**



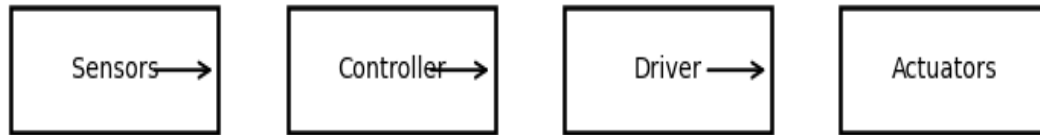
- Robotics combines mechanical engineering, electronics, electrical engineering, control systems and programming. Robots use sensors to collect information, controllers to process it and actuators to create physical action.
- Robotics is used in education, manufacturing, inspection, logistics, healthcare, agriculture and research.
- This report follows the beginner-friendly progression of the uploaded Arduino tutorial: introduction, components, sensors, programming, projects, troubleshooting and conclusion.

• What Is a Robot?



- A robot is a programmable machine designed to perform physical tasks automatically or semi-automatically.
- A robot combines sensing, computation and actuation. Examples include mobile robots, robotic arms, drones, warehouse robots and educational robots.
- Autonomy can range from remote control to systems that make decisions from sensor information.
- A robot is a machine that can:
 - ✓ Sense its surroundings
 - ✓ Process information
 - ✓ Make decisions according to its program
 - ✓ Perform physical actions
 - ✓ Repeat tasks with accuracy
 - ✓ Example: A line-following robot senses a black line using IR sensors and controls its motors to follow the path.

- **Basic Robotic System**



Basic Robotic System

- A basic robot can be represented as Sensors → Controller → Actuators → Environment, with feedback returning to the controller.
- Sensors observe the environment or robot state. The controller processes measurements and actuators create motion.
- This feedback concept is used in simple obstacle avoidance as well as advanced autonomous robots.

1. Sensors

Sensors collect information from the surroundings.

Examples:

Ultrasonic sensor – detects distance

IR sensor – detects objects/lines

Temperature sensor – measures temperature

Light sensor – detects light

2. Controller

The controller is the brain of the robot. It receives sensor data, processes it according to the program, and sends commands to other components.

Examples: Arduino, Raspberry Pi, microcontrollers.

3. Motor Driver

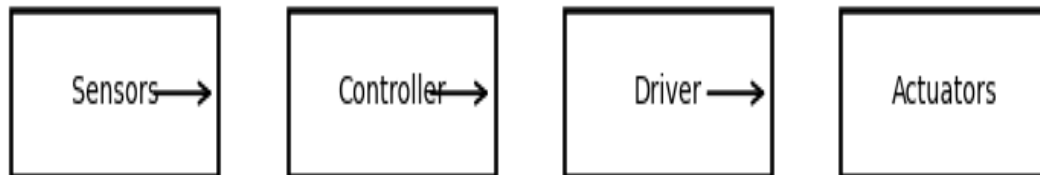
A motor driver connects the controller to the motors. It provides the required current and allows the controller to control the speed and direction of motors.

4. Actuators

Actuators convert electrical signals into physical movement.

Examples: DC motors, servo motors, stepper motors, and pneumatic actuators.

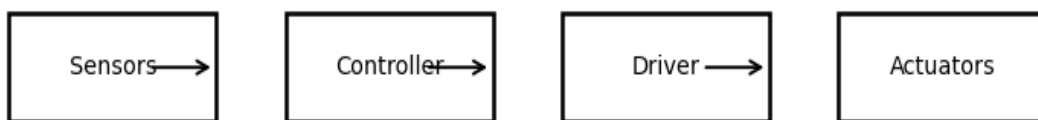
- **Main Parts of a Robot**



Main Parts of a Robot

- Main parts are mechanical structure, controller, sensors, actuators, power supply and communication interfaces.
- The mechanical structure supports the robot; sensors measure conditions; the controller runs the program; actuators create movement; power supplies provide energy.

- **Robot Controller**



Robot Controller

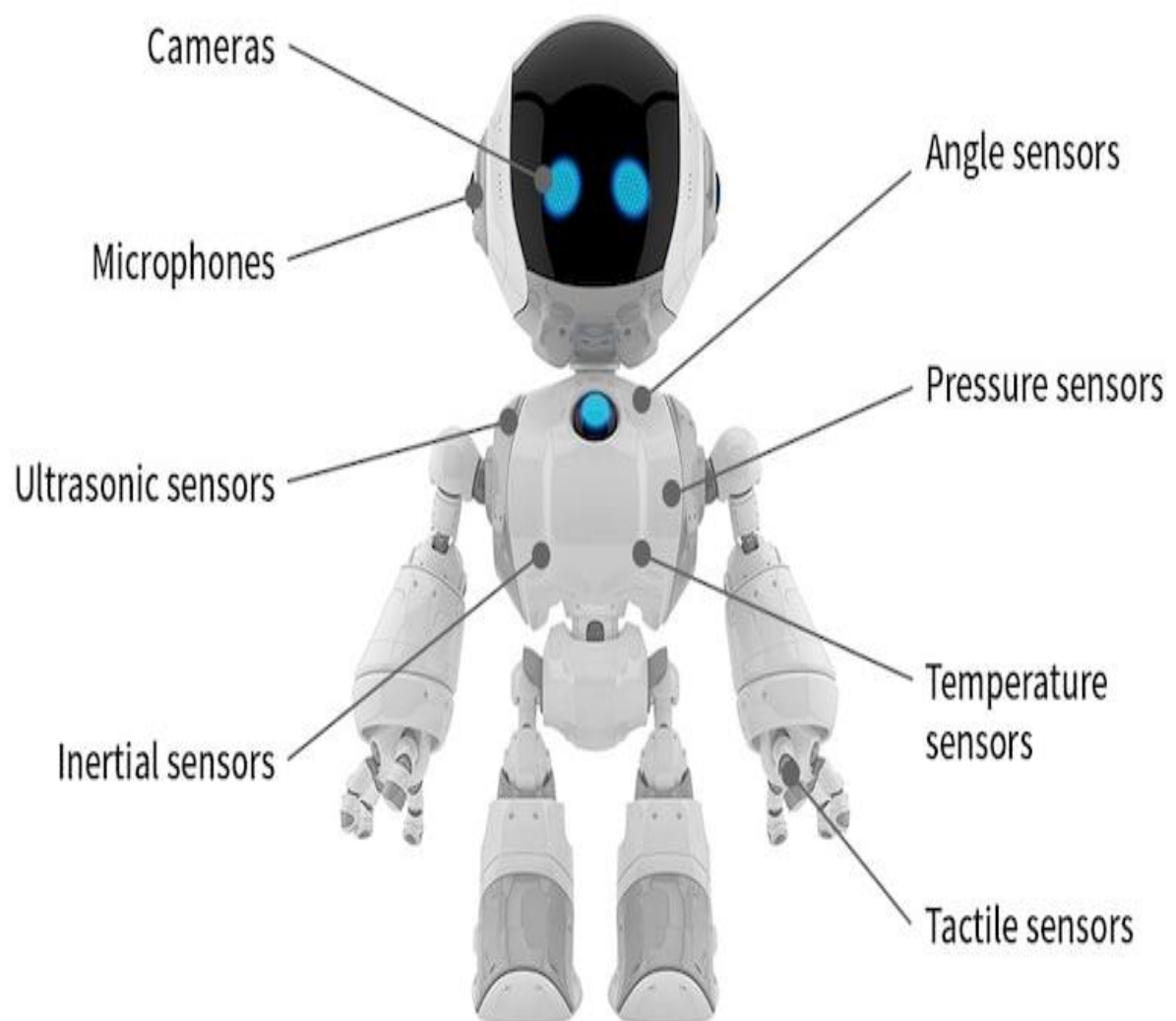
- The controller is the decision-making unit. Educational robots often use Arduino boards; advanced systems can use PLCs, single-board computers or dedicated motion controllers.
- The controller reads sensors, runs algorithms and commands motors or other outputs.

- **Arduino in Robotics**



- The uploaded reference describes Arduino as an open-source electronics platform with a programmable microcontroller board and Arduino IDE. It can accept sensor inputs and control outputs such as motors, LEDs, buzzers and displays.
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- Arduino Uno is widely used by beginners and students and provides digital pins, analog inputs and PWM-enabled pins. □ filecite□turn0file0□L27-L30□
- In robotics, Arduino can read sensors and control a motor driver.

- **Sensors Used in Robots**



1. Camera sensors

Cameras act like the eyes of the robot. They capture images of the surroundings. Computer-vision software can then use those images for tasks such as object recognition, navigation and localization.

Example: A service robot uses a camera to recognize a person or identify an object on a table.

2. Microphone sensors

Microphones act like the ears of the robot. They detect sound and can be used for voice commands, sound localization and interaction with humans.

Example: A robot hears “move forward” and its speech-processing system converts the command into an action.

3. Ultrasonic sensors

An ultrasonic sensor sends a high-frequency sound pulse and measures the time taken for the echo to return. This allows the robot to estimate the distance to an object.

Example: If an obstacle is detected in front of a robot, it can stop or change direction.

4. Inertial sensors

Inertial sensors such as accelerometers and gyroscopes measure the robot's acceleration and rotation. They help the robot determine its movement, orientation and balance.

Example: A humanoid robot uses an IMU to detect that its body is tilting and adjust its movement to maintain balance.

5. Angle sensors

Angle sensors measure the position or angle of robot joints. They are commonly implemented using encoders or similar position sensors.

Example: In a robotic arm, an angle sensor tells the controller whether the elbow joint is at 30°, 60°, etc.

6. Pressure sensors

Pressure sensors detect the amount of force or pressure applied to a part of the robot. They are useful for controlling how strongly a robot holds or pushes an object.

Example: A robotic hand can use pressure feedback to hold an egg without applying excessive force.

7. Temperature sensors

Temperature sensors monitor the temperature of components such as motors, batteries and electronics. This can help detect overheating and protect the robot. Example: If a motor becomes too hot, the controller can reduce its operation or stop it.

8. Tactile sensors

Tactile sensors are similar to the robot's sense of touch. They detect physical contact and can provide information about pressure, force or contact conditions. Example: When a robot touches an object with its finger, tactile sensors can tell the robot that contact has occurred.

• Robot Power Supply

A power supply is an important part of a robot. It provides the required electrical energy to the microcontroller, sensors, motors, and other electronic components.

1. Purpose of Power Supply

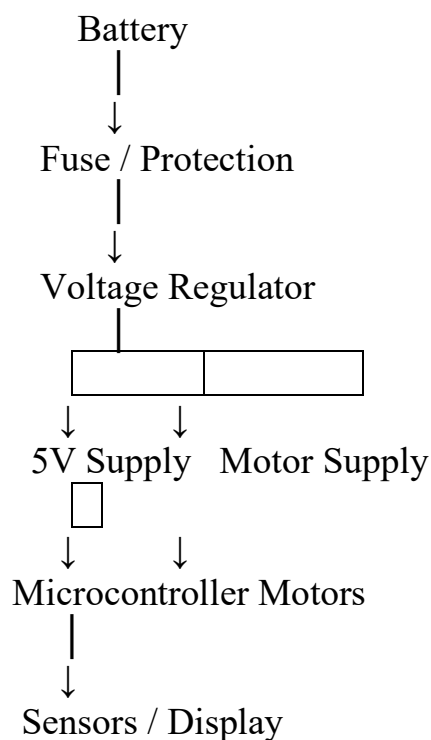
The main functions are:

- Provides electrical power to all robot components.
- Converts the available battery voltage into the required voltage levels.
- Maintains stable voltage for electronic circuits.
- Provides sufficient current for motors and actuators.
- Protects the circuit from overvoltage and short circuits.

2. Common Power Sources

- AA/AAA batteries
 - ✓ 1.2–1.5 V each
 - ✓ Small robots
- Li-ion battery
 - ✓ 3.7 V/cell
 - ✓ Mobile robots
- Li-Po battery
 - ✓ 3.7 V/cell
 - ✓ Drones, high-performance robots
- 9 V battery
 - ✓ 9 V
 - ✓ Small low-power circuits
- Lead-acid battery
 - ✓ 6/12 V
 - ✓ Larger robots

3. Basic Robot Power Supply Block Diagram



4. Voltage Regulation

Different robot components may require different voltages.

For example:

Microcontroller: 3.3 V or 5 V

Sensors: 3.3 V or 5 V

Motor driver: 5 V logic supply

DC motors: commonly 6 V, 9 V, 12 V, etc.

A voltage regulator or DC-DC converter is used to obtain the required voltage from the battery.

5. Battery Selection

Battery selection depends mainly on:

Voltage + Current capacity + Operating time + Weight

For a mobile robot, rechargeable Li-ion or Li-Po batteries are commonly used because they provide good energy capacity with relatively low weight.

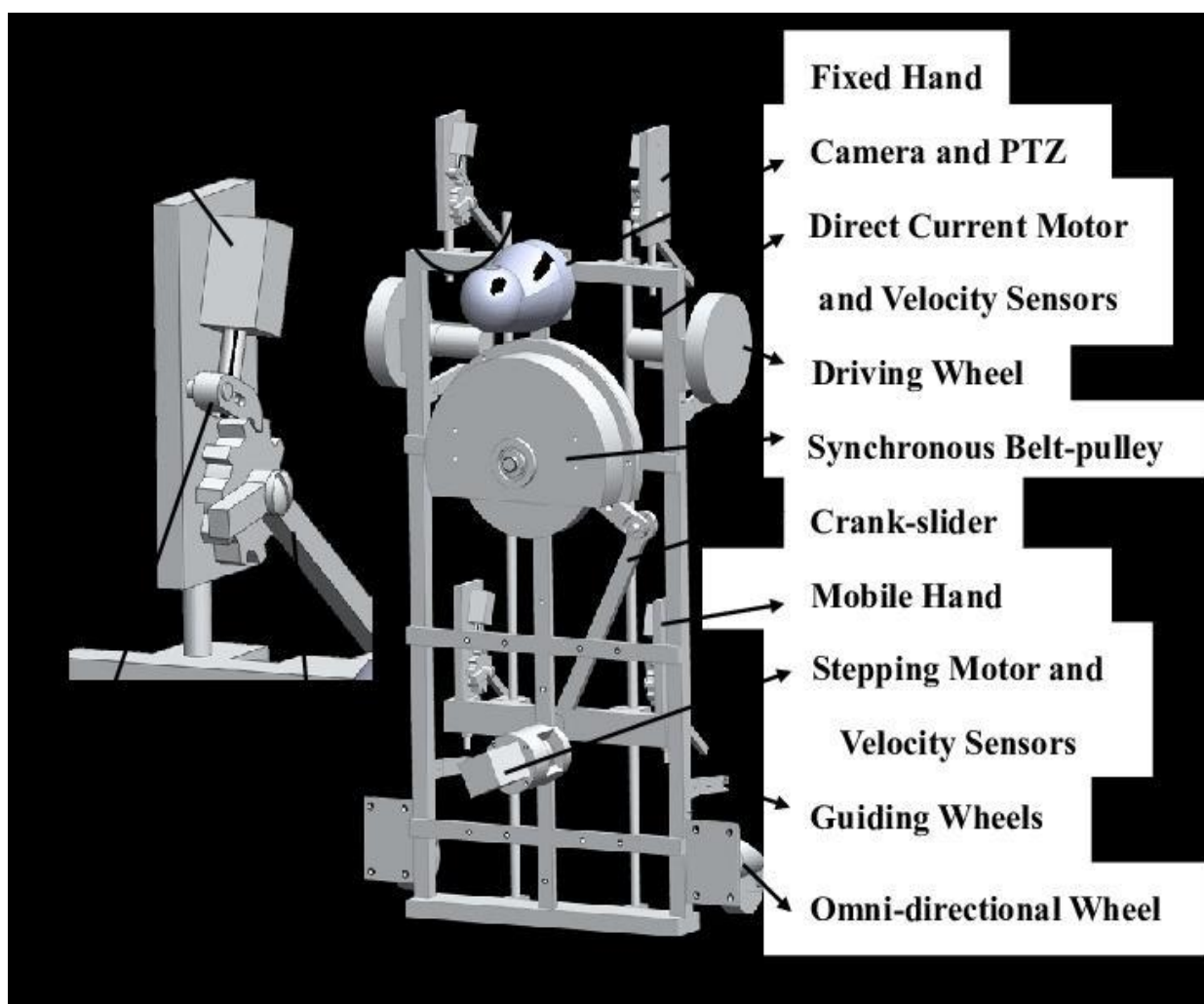
6. Important Point

Motors can draw much more current than sensors or microcontrollers. Therefore, it is generally better to provide the motor power through a suitable motor driver and adequate battery, while using a regulated supply for sensitive electronics.

7. Advantages of a Good Power Supply

- ✓ Stable robot operation
- ✓ Longer battery life
- ✓ Prevents component damage
- ✓ Reduces unexpected resets
- ✓ Provides sufficient motor performance
- ✓ Improves overall reliability

- **Robot Mechanical Structure**



- The mechanical structure supports components and transfers actuator motion.

- Mobile robots may use differential drive or four-wheel arrangements. Robotic arms use links and joints.
- Design should consider weight, center of mass, traction, payload, strength and maintenance.

- **Robot Types**

- Robots can be classified as mobile robots, industrial arms, humanoids, aerial robots, underwater robots, educational robots and service robots.
- Classification can also be based on application, mobility, control method and autonomy.

- **Mobile Robots**



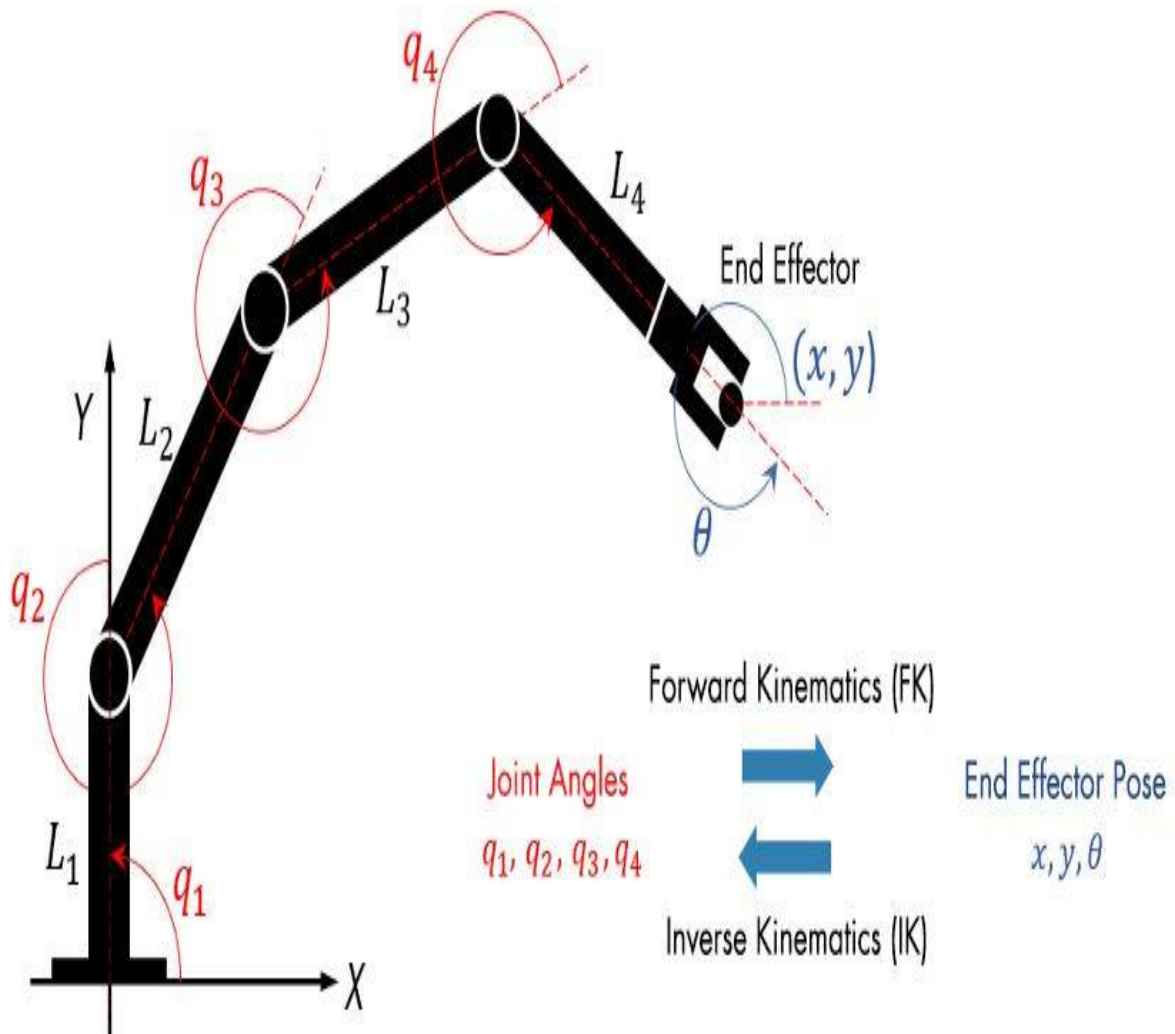
- Mobile robots use wheels, tracks or other mechanisms to move.
- Differential-drive robots are common educational projects because two independently controlled wheels can provide forward, reverse and turning motion.
- Applications include logistics, cleaning, delivery, inspection and research.

- **Robotic Arm**



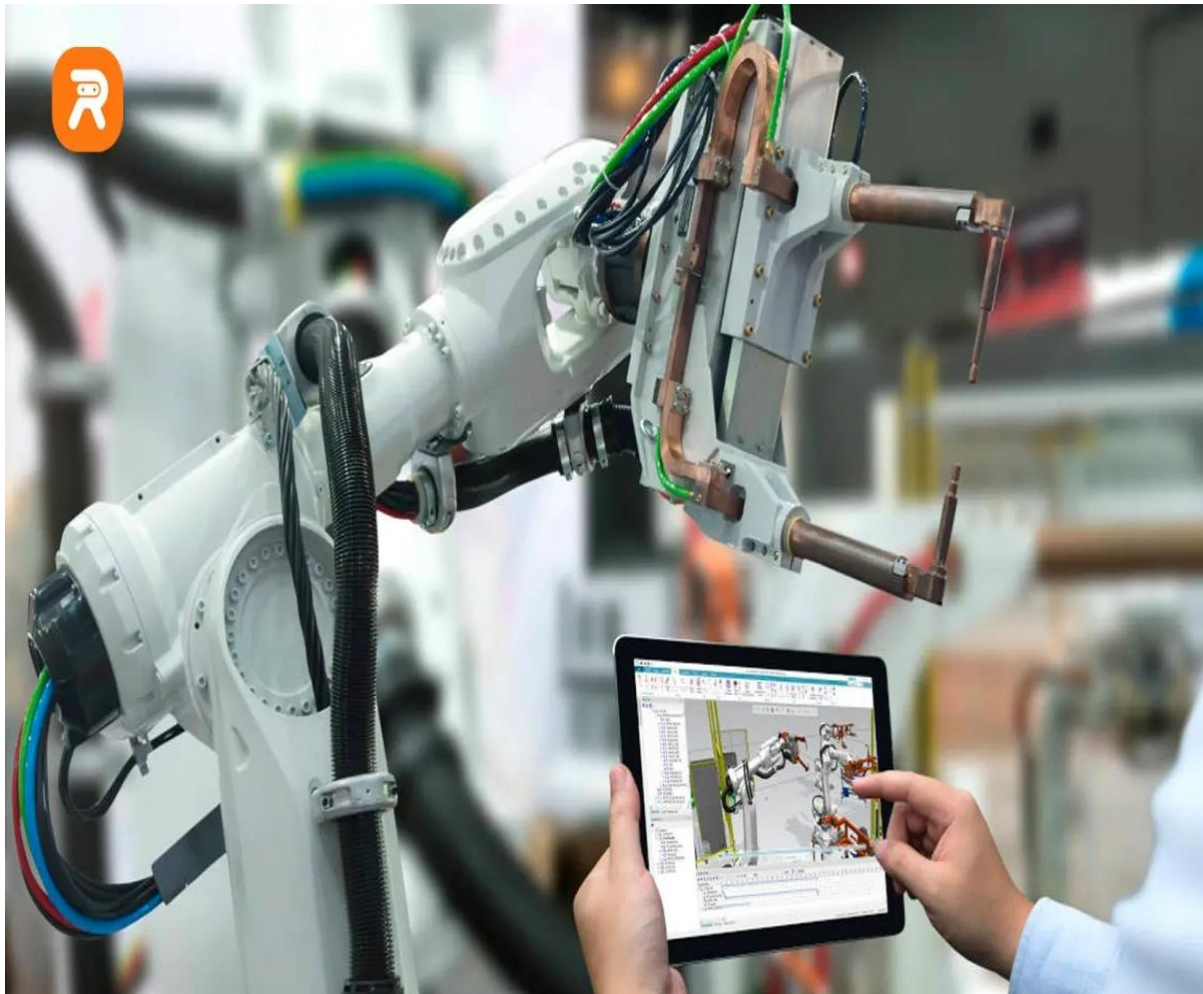
- A robotic arm consists of rigid links connected by joints. Joint actuators position an end-effector.
- End-effectors can be grippers, suction tools, welding tools, drills or cameras.
- Simple educational arms often use servo motors.

- **Robot Kinematics**



- Kinematics describes robot motion without concentrating on forces.
- Forward kinematics finds end-effector position from joint variables. Inverse kinematics finds joint variables for a desired position.
- Kinematic models are important for robotic-arm positioning.

- **Robot Control**



- Control makes a robot follow desired behavior.
- Open-loop control does not measure the result; closed-loop control uses feedback.
- Motor speed control with encoder feedback is a common example. PID is a widely used feedback method.

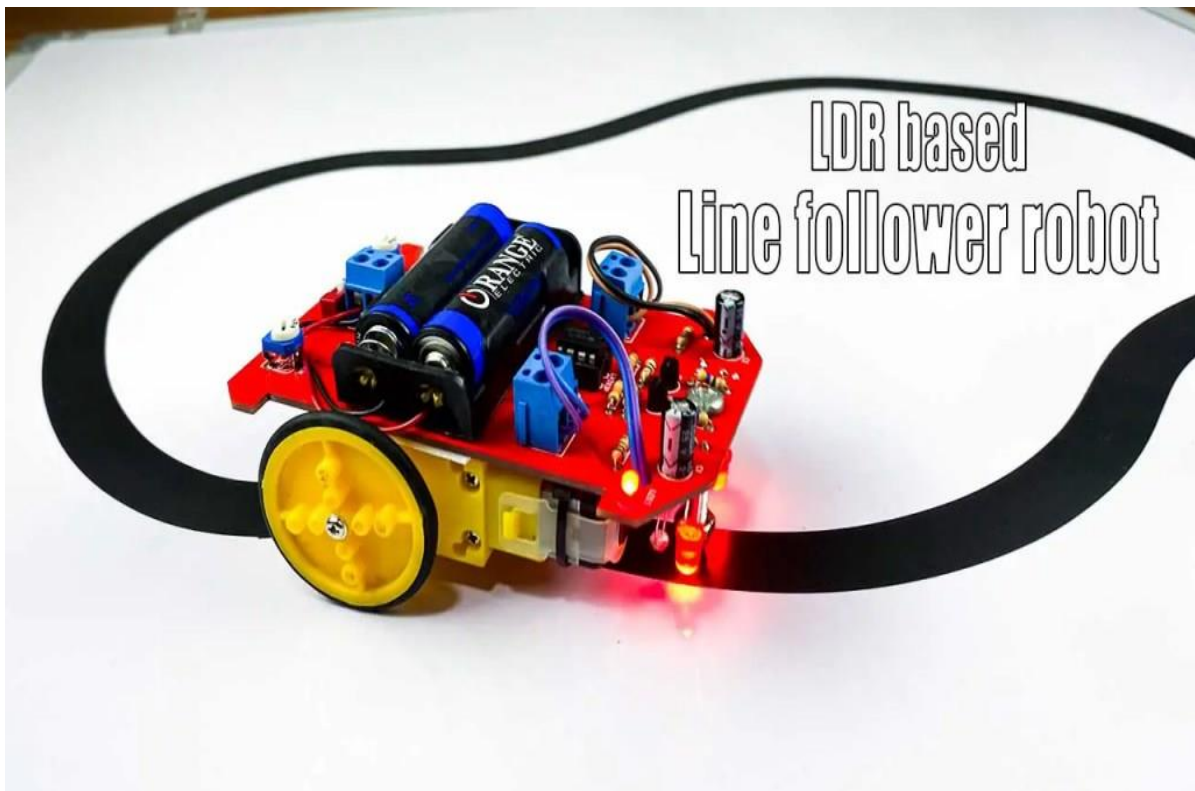
- **Programming a Robot**

- A robot program commonly follows: initialize → read sensors → process → decide → command actuators → repeat.
- The uploaded Arduino reference states that setup() performs one-time initialization and loop() contains repeated instructions.
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- Good programs use clear variables, functions, safe default states and serial messages during testing.

- **Robot Communication**

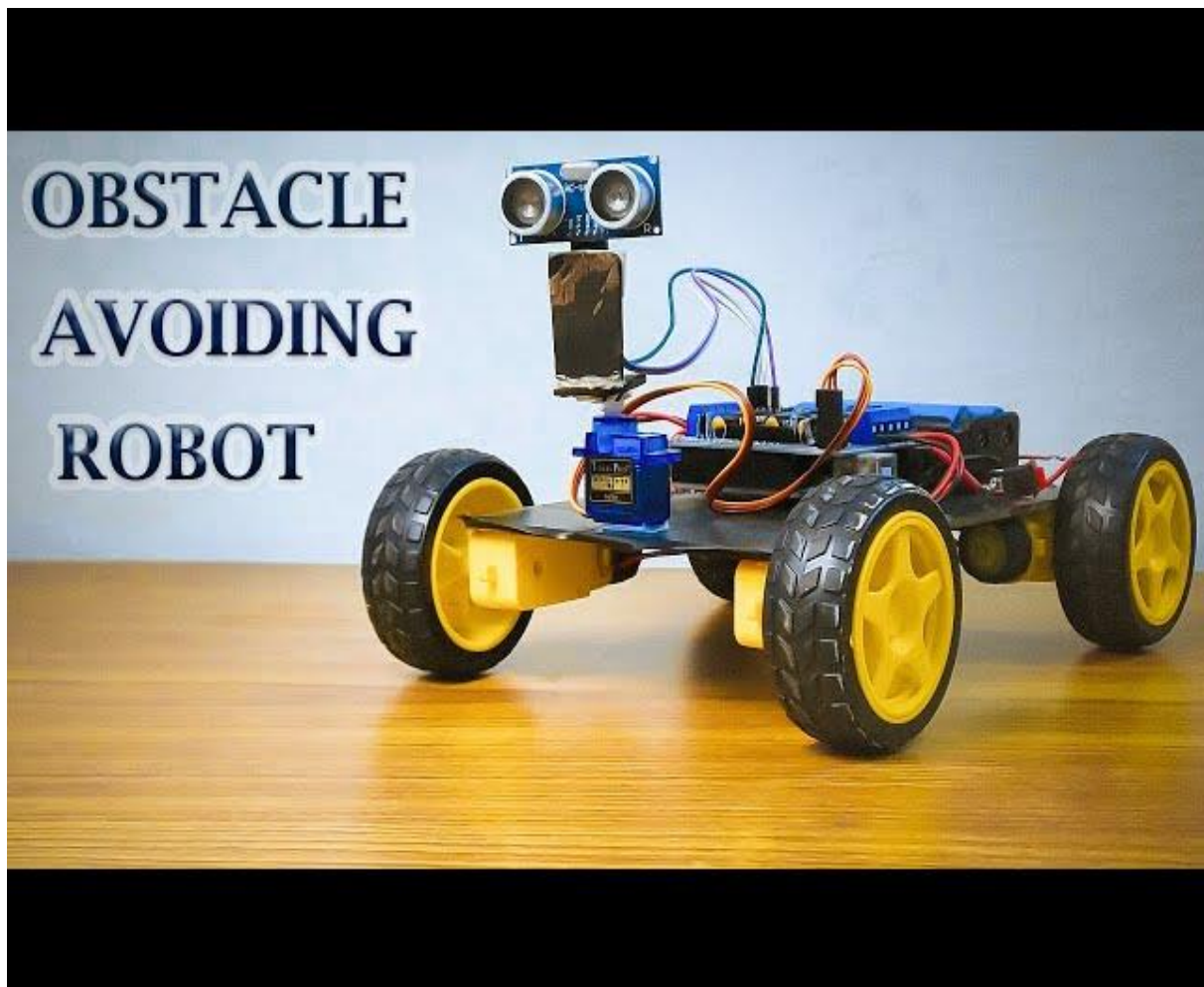
- Robots communicate internally between sensors, controllers and drivers and externally with computers or networks.
- Common interfaces include UART/serial, I2C and SPI; wireless systems may use Bluetooth or Wi-Fi.
- Communication design considers speed, distance, reliability and electrical compatibility.

- **Line Following Robot**



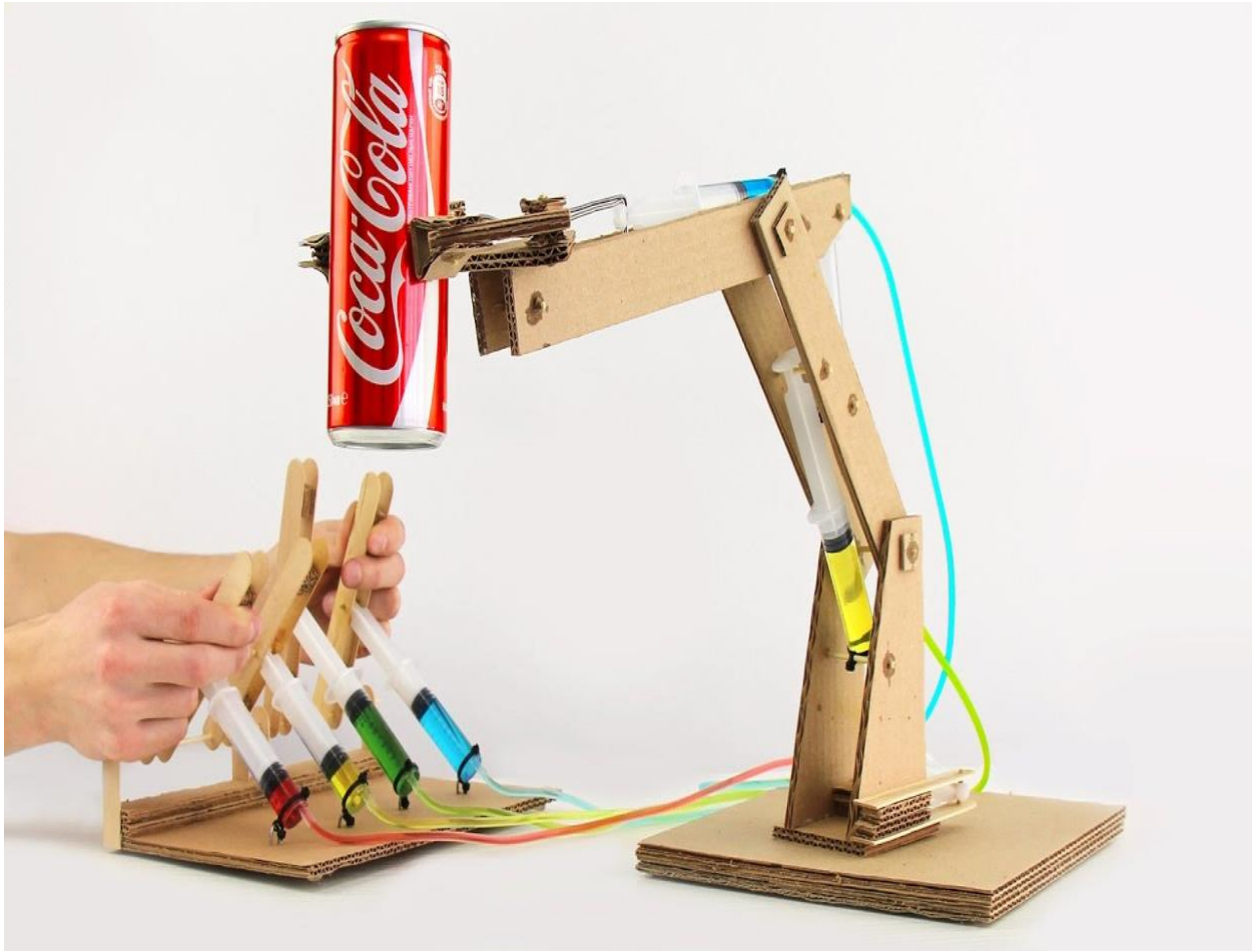
- A line-following robot uses IR sensors to detect a path and adjusts motor speeds.
- A simple controller compares left and right sensor states and corrects the robot's direction.
- This project demonstrates sensing, decision-making, actuation and feedback.

- **Obstacle Avoiding Robot**



- An obstacle-avoiding robot measures distance and changes direction when an object is too close.
- A simple algorithm is: move forward above a threshold; stop and turn at the threshold; measure again.
- The threshold should be tested for the robot's speed and stopping distance.

- **Robotic Arm Project**



- A small robotic arm can use an Arduino-compatible controller, servo motors and a gripper.
- A sequence can be pickup → close gripper → move → open gripper → return.
- Joint limits and payload limits should be respected.

- **Industrial Robotics**



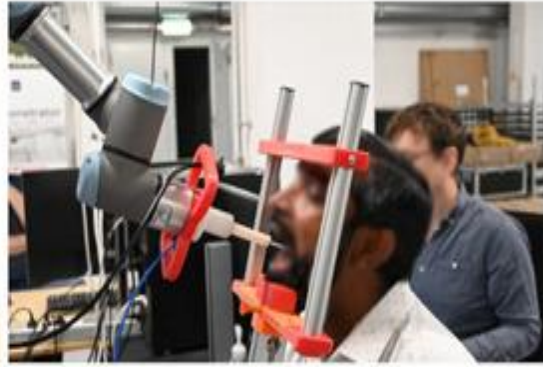
- Industrial robots are used for welding, painting, assembly, material handling, packaging and inspection.
- An industrial system includes a manipulator, controller, end-effector, sensors, fixtures and safety systems.
- Maintenance, operator training and risk assessment are important.

- **Service, Medical and Agricultural Robots**





(a)



(b)



(c)



(d)

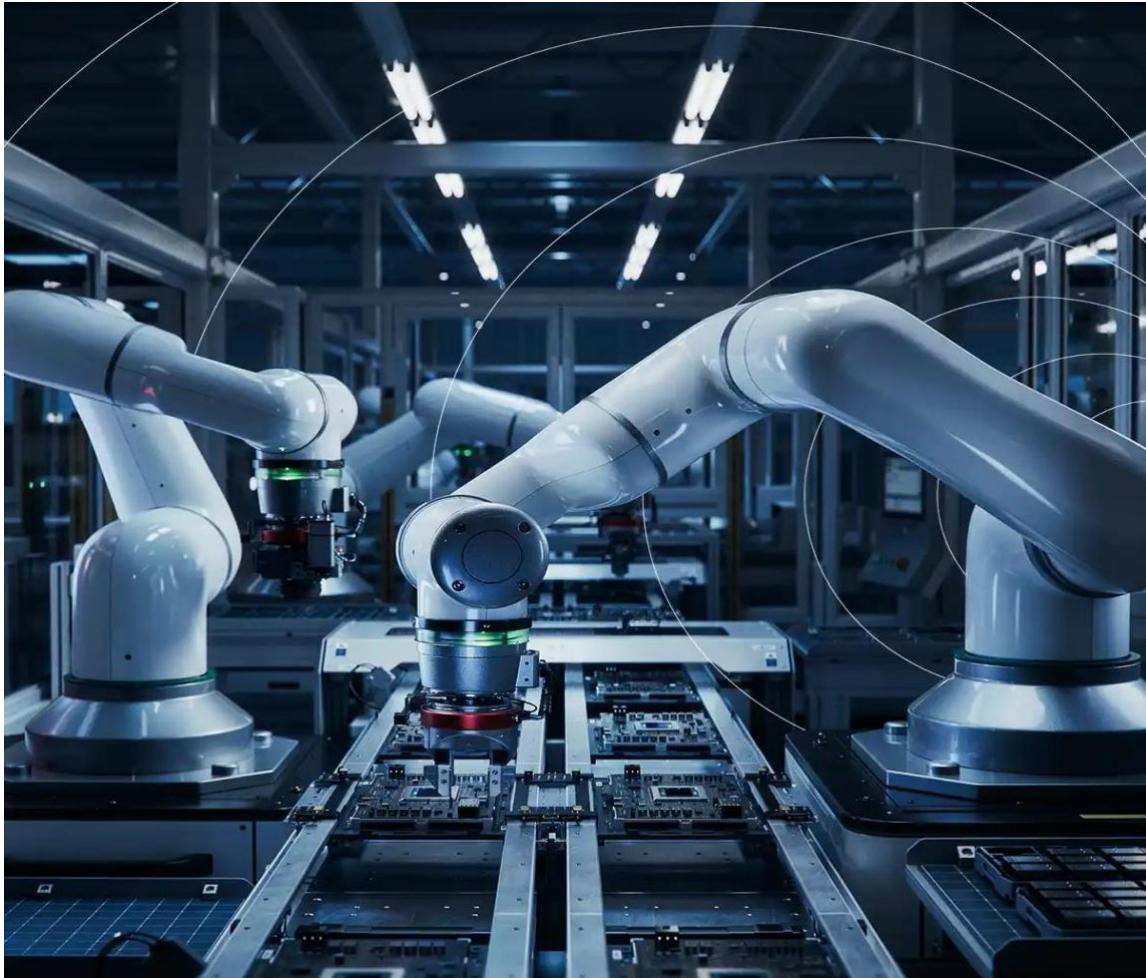
- Service robots may assist with cleaning, delivery, logistics or inspection. Medical systems can support selected clinical or rehabilitation tasks. Agricultural robots can monitor crops and support field operations.
- These robots must be designed for their specific environment and safety requirements.

- **AI and Robotics**



- AI can help robots interpret images, classify objects, estimate states, plan actions or learn patterns.
- Computer vision can identify objects or lanes. Machine learning can process sensor data while conventional control can manage low-level motor behavior.
- AI still depends on reliable sensing, control, testing and safety.

- **IoT and Connected Robotics**



- IoT-enabled robots can exchange sensor data with local or cloud systems.
- A connected inspection robot may send battery, temperature, location and image information to a dashboard.
- Connectivity requires attention to network reliability, authentication, data protection and safe behavior during communication loss.

- **Advantages, Limitations and Safety**

- Advantages include automation, repeatability, operation in difficult environments and programmable behavior.
- Limitations include cost, maintenance, battery limits, sensor errors and programming complexity.
- Safety requires appropriate shutdown/emergency behavior, guarded moving parts, electrical protection and testing before operation around people.

- **Future of Robotics and Conclusion**

- Robotics is moving toward collaborative robots, autonomous navigation, intelligent perception and connected systems.
- A useful learning path is electronics → controller programming → sensors → motors → line follower → obstacle avoidance → robotic arm → computer vision and autonomous navigation.
- Conclusion: robotics integrates hardware and software through the sensing–decision–actuation cycle.